cryptographic mechanisms that provide forward secrecy as discussed in section 12.2. In that section, we looked at public key mechanisms for achieving forward secrecy, but you can also achieve this security goal using purely symmetric cryptography through a technique known as ratcheting.

> **DEFINITION** ***Ratcheting*** in cryptography is a technique for replacing a symmetric key periodically to ensure forward secrecy. The new key is derived from the old key using a one-way function, known as a ***ratchet***, because it only moves in one direction. It's impossible to derive an old key from the new key so previous conversations are secure even if the new key is compromised.

There are several ways to derive the new key from the old one. For example, you can derive the new key using HKDF with a fixed context string as in the following example:

```
var newKey = HKDF.expand(oldKey, "iot-key-ratchet", 32,
"HMAC");
```

> **TIP** It is best practice to use HKDF to derive two (or more) keys: one is used for HKDF only, to derive the next ratchet key, while the other is used for encryption or authentication. The ratchet key is sometimes called a *chain key* or *chaining key*.

If the key is not used for HMAC, but instead used for encryption using AES or another algorithm then you can reserve a particular nonce or IV value to be used for the